

Project Title
Large Prime Number Validation Using Miller–Rabin Algorithm

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents a web-based service for validating extremely large integers using the Miller–Rabin Primality Test, a randomized probabilistic algorithm. Traditional trial-division methods become inefficient for very large numbers, whereas Miller–Rabin can determine whether a number is likely prime or composite efficiently. The system accepts a large integer as input and performs multiple randomized rounds to test its primality. If a witness proves compositeness, the number is reported as composite; otherwise, it is reported as probably prime with a high confidence level based on the number of rounds performed. The application also displays the execution time, result, and confidence level, allowing users to observe the efficiency of the probabilistic approach. This project demonstrates probabilistic algorithm design and its practical application to large-number primality testing.

Signature of the Guide